Highlights

Tree Ensembles and Tabular Transformers for Financial Fraud Detection: Imbalance Sensitivity, Controlled Transfer and Interpretation Boundaries

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- Boosted trees lead transfer point estimates on five filtered BAF cohorts
- FT-Transformer loses about 35% average precision with primary SMOTE
- A categorical-aware control does not recover the transformer baseline
- SHAP set agreement does not establish rank stability or causality
- Diffuse final-layer attention is a separate descriptive diagnostic

Tree Ensembles and Tabular Transformers for Financial Fraud Detection: Imbalance Sensitivity, Controlled Transfer and Interpretation Boundaries

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ABSTRACT

Close in-domain scores do not establish that fraud models respond similarly to imbalance interventions or distribution shift. We compare classical models and a custom FT-Transformer through imbalance sensitivity, frozen Base-to-Variant transfer, compatible SHAP analysis and a separate attention diagnostic. The ULB experiment separates distinct predictor profiles before splitting. BAF transfer excludes exact matches to Base development profiles from each Variant test cohort, retaining 177,526–178,828 observations under a common 30-predictor projection. Preprocessing and thresholds are selected within development data. On BAF Base, CatBoost, FT-Transformer and LightGBM baseline average precision is 0.1796, 0.1769 and 0.1768, respectively. Primary SMOTE reduces FT-Transformer average precision by approximately 35%, compared with 14% for CatBoost. A predefined categorical-aware FT-Transformer control reaches 0.1157, not the 0.1769 baseline; the comparison does not isolate an architectural cause. LightGBM leads transferred average precision on Variant I and CatBoost on Variants II–V. Every supervised model has lower transferred F_2 than on Base under its frozen threshold. The boosted models share a top-10 SHAP feature set, and the fixed LightGBM model retains its members across the evaluated BAF cohorts while contribution magnitudes vary. These are not identical rankings or guaranteed explanation validity. FT-Transformer final-layer, head-averaged attention is diffuse (normalised entropy 0.985), a descriptive result rather than evidence of causality or an information ceiling. Together, these analyses support evidence-bounded model selection while keeping representation, operating-point and interpretation limitations explicit.

1. Introduction

Financial fraud detection combines rare positive cases with an operational trade-off between missed fraud and false alerts (Ryman-Tubb, Krause and Garn, 2018; Hilal, Gadsden and Yawney, 2022). Choosing a model therefore requires more than comparing one in-domain score. A useful evaluation also asks how imbalance interventions affect the implemented pipeline, how a frozen model behaves on changed populations, and what its inspected explanations support.

Gradient-boosted trees remain strong tabular baselines (Ke, Meng, Finley, Wang, Chen, Ma, Ye and Liu, 2017; Prokhorenkova, Gusev, Vorobev, Dorogush and Gulin, 2018; Shwartz-Ziv and Armon, 2022), while attention-based models such as FT-Transformer offer a different representation of feature interactions (Gorishniy, Rubachev, Khrulkov and Babenko, 2021). Benchmark and survey evidence motivates dataset-dependent comparison rather than a universal architectural ordering (Grinsztajn, Oyallon and Varoquaux, 2022; Borisov, Leemann, Seßler, Haug, Pawelczyk and Kasneci, 2024). The companion article (Caixeiro, Abbasi and Martins, 2026) examines baseline performance and validation-derived decision thresholds. The present article uses the same pinned evidence but focuses on sensitivity, controlled transfer and the boundaries of interpretation.

We make four contributions:

C1. **Representation-sensitive imbalance comparison.** FT-Transformer and CatBoost are compared across seven predefined strategies on ULB and BAF, using six effective conditions in intervention summaries, with categorical-aware BAF controls reported separately.

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C2. **Controlled transfer on audited cohorts.** Base-fitted models and thresholds are frozen before evaluation on five BAF Variant cohorts excluding exact common-predictor matches to Base DEV.

C3. **Compatible SHAP set comparisons.** Linear and boosted-tree explanations are compared at the original-feature level; the fixed Base-trained LightGBM is also inspected on retained Variants.

C4. **Separate attention diagnostics.** Final-layer, head-averaged FT-Transformer attention is described without treating it as faithful attribution or a test of a dataset-imposed ceiling.

The contribution is an integrated, traceable evaluation, not a causal account of architectural superiority. It distinguishes ranking performance from threshold-dependent performance, representation effects from architecture, and global feature-set agreement from ranking or causal stability. Section 2 reviews the methods; Section 3 defines the protocol; Section 4 presents the evidence; Sections 5 and 6 discuss its implications.

The research code and article-specific result summaries are provided in a dedicated repository at `https://github.com/Applied-Intelligence-Hub/transformers-vs-ml-fraud-detection`. Dataset access instructions and the scope of the released evidence are described in the Data Availability statement.

2. Related Work

2.1. Tabular modelling and imbalance interventions

LightGBM and CatBoost are distinct gradient-boosting implementations (Ke et al., 2017; Prokhorenkova et al., 2018); Random Forest combines randomised trees (Breiman, 2001). FT-Transformer tokenises numerical and categorical inputs before transformer processing (Gorishniy et al., 2021). The comparative literature identifies strong tree baselines and dependence on datasets and preprocessing choices (Shwartz-Ziv and Armon, 2022; Grinsztajn et al., 2022; Borisov et al., 2024). Those findings motivate comparison but do not predict a universal winner under fraud-specific interventions.

SMOTE synthesises minority observations by interpolation (Chawla, Bowyer, Hall and Kegelmeyer, 2002); cleaning combinations add Tomek-link or edited-nearest-neighbour stages (Batista, Prati and Monard, 2004). The representation on which interpolation operates matters. Interpolating one-hot columns or ordinal category codes differs from preserving nominal category membership. We report the executed preprocessing and sampling paths rather than attributing all differences to architecture.

2.2. Controlled distribution transfer

The BAF suite provides a synthetic framework for examining imbalance, bias and distribution changes (Jesus, Pombal, Alves, Cruz, Saleiro, Ribeiro, Gama and Bizarro, 2022b,a). Its Variants share a generating process with Base, making them useful controlled stress tests but not independent institutional deployment studies. Freezing Base preprocessing, models and thresholds avoids Variant-specific adaptation. Exact-profile overlap must also be checked: a separate CSV need not contain profiles unseen during Base development.

Random Forest randomisation is a modelling design (Breiman, 2001), not a theorem that bagged trees transfer better than boosted trees or transformers. A change in average precision may reflect both score ordering and the evaluated population, including prevalence. We therefore report retained cohorts, multiple metrics and the fixed operating rule.

2.3. SHAP and attention have different scopes

SHAP decomposes model output relative to an explainer reference (Lundberg and Lee, 2017; Lundberg, Erion, Chen, DeGrave, Prutkin, Nair, Katz, Himmelfarb, Bansal and Lee, 2020). It describes the fitted model, and depends on feature dependence, background choice and grouping (Molnar, 2025). Jaccard similarity of top-k sets measures membership overlap:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

It does not compare ranking order, attribution magnitudes or causal effects.

Attention weights describe allocation within a network computation. A named feature token does not make its attention weight the feature's effect on the final output. Head averaging may conceal structure; residual paths and feed-forward blocks are outside the inspected matrix. SHAP and attention are therefore separate analyses here, not independent measurements validating a shared causal construct.

3. Experimental Setup

3.1. Datasets and evaluated populations

Raw ULB 2013 contains 284,807 transactions and 492 frauds (Machine Learning Group, Université Libre de Bruxelles, 2016; Dal Pozzolo, Caelen, Johnson and Bontempi, 2015). Exact predictor-profile deduplication before splitting retains 283,726 profiles and 473 frauds. A stratified seed-42 split assigns 226,980 profiles (378 frauds) to development (DEV) and 56,746 (95 frauds) to TEST. Predictors exclude the target; the 28 PCA components, transaction time and amount remain (30 predictors). Estimates concern distinct observed profiles, not the original transaction-frequency population. Customer independence is not inferred.

BAF Base contains 1,000,000 applications; DEV contains 800,000 and TEST 200,000, including 2,206 TEST frauds. We use 30 common predictors: 25 treated as numerical and five nominal categorical variables. The target and `month` are excluded; months are pooled. The extra `x1`/`x2` variables in Variants III/V are excluded to preserve the same schema.

From each original 200,000-row Variant TEST partition, every common-30 predictor profile matching a Base DEV profile is excluded. Numeric canonicalisation and element-wise confirmation supplement hash candidate matching; exclusions do not use Variant labels. Table 1 gives the populations shared by transfer and Variant SHAP. These

Table 1
BAF transfer cohorts excluding exact matches to Base DEV. Parentheses give excluded fraud cases.

Cohort	Original	Excluded	Retained	Frauds	Prevalence (%)
Base	200,000	0 (0)	200,000	2,206	1.1030
Variant I	200,000	22,474 (262)	177,526	1,944	1.0951
Variant II	200,000	22,469 (250)	177,531	1,956	1.1018
Variant III	200,000	21,282 (245)	178,718	1,961	1.0973
Variant IV	200,000	21,350 (246)	178,650	1,960	1.0971
Variant V	200,000	21,172 (228)	178,828	1,978	1.1061

safeguards establish exact-profile separation from Base DEV, not entity independence, separation from Base TEST or mutual independence among Variants. Filtering changes the population; the common projection does not isolate full separability interventions involving `x1`/`x2`.

3.2. Preprocessing, models and selection

The five supervised families are Logistic Regression (LR), Random Forest (RF), LightGBM (LGBM), CatBoost and a custom FT-Transformer. One-Class SVM (OCSVM) is a separate reference, fitted only on legitimate training observations; validation labels still select its threshold. It uses an RBF kernel, $\nu = 0.01$ and `gamma=scale`, without Optuna tuning.

Numerical imputation and standardisation are fitted only within the relevant training partition. Classical pipelines one-hot encode nominal categories, including CatBoost; native CatBoost categorical processing is not evaluated. FT-Transformer uses learned numerical tokens and embeddings of integer category codes. This custom implementation is one representative of tabular transformers, not an assessment of every FT-Transformer implementation.

The primary BAF policy preserves documented numerical absence codes (Jesus et al., 2022a). Actual `NaN` values are mean-imputed, but coded absence enters standardisation as a numerical magnitude. A predefined sensitivity converts only documented absence codes to `NaN`, adds indicators and applies training-fitted mean imputation. It holds baseline hyperparameters fixed; the primary policy is not chosen retrospectively using TEST.

Classical supervised baselines use 50 attempted Optuna TPE trials (Akiba, Sano, Yanase, Ohta and Koyama, 2019), with pruning and five-fold stratified inner validation. Hyperparameters maximise validation AP and are reused across imbalance strategies. FT-Transformer uses one stratified 80/20 holdout within DEV instead of five-fold validation. ULB repeats searches on the distinct-profile partition; BAF retains its previously selected hyperparameters. Frozen classical BAF final models are preserved while validation evidence is recovered. BAF FT-Transformer follows the deterministic validation/final-training schedule with retained hyperparameters, without a new BAF search.

For the ULB search, FT-Transformer trial initialisation uses seed 42+trial number. A seed-42 validation refit selects checkpoint, threshold and epoch count using validation AP. A freshly initialised seed-42 model then fits all DEV for that epoch count. AdamW and cosine scheduling retain a 200-epoch horizon. The BAF baseline has 64-dimensional

tokens, two transformer blocks and eight heads. Validation scores, indices, preprocessors and checkpoints identify the executed path. The two validation designs protect TEST but do not estimate the same selection variability.

3.3. Imbalance and representation controls

The primary comparison uses None, random undersampling (RUS), random oversampling (ROS), SMOTE, SMOTE+Tomek, SMOTEENN and Class Weights. Samplers act only on training data, targeting class balance. Baseline hyperparameters are reused rather than retuned for every strategy. All seven strategies remain displayed. The observed SMOTE+Tomek duplicate is omitted only from between-strategy summaries, which use None, RUS, ROS, SMOTE, SMOTEENN and Weights with population standard deviation ($ddof = 0$).

Primary classical SMOTE interpolates standardised numerical and one-hot categorical columns. Primary FT-Transformer SMOTE interpolates numerical values and ordinal category codes, subsequently converting synthetic category values to integer embedding indices. It does not interpolate learned category embeddings. Neither path is categorical-aware. Separate fixed-hyperparameter SMOTE-NC controls for BAF CatBoost and FT-Transformer preserve the five nominal categories. Numerical binary variables remain in the interpolated numerical block, so not all discrete constraints are enforced. Controls are outside the seven-strategy matrix.

3.4. Metrics, thresholds and uncertainty

PR-AUC denotes average precision (AP), the primary ranking summary; ROC-AUC is complementary (Saito and Rehmsmeier, 2015). AP groups exactly equal scores and is invariant to their row order; it is not a position-wise AP after arbitrary tie-breaking. Accompanying top-k derivatives use the analytic expectation under uniform selection within an exact boundary tie, preserving the original budget (McSherry and Najork, 2008). The operational rule maximises validation F_2:

$$F_2 = \frac{5TP}{5TP + 4FN + FP}.$$

For classical models, the final threshold is the median of five fold-selected thresholds. FT-Transformer uses saved internal validation scores. Final models fit all DEV; thresholds are frozen before TEST. No TEST fitting, probability calibration or threshold selection occurs. Threshold selection is not probability calibration. The median reduces sensitivity to extreme fold cuts without guaranteeing the best cut for the final fit. FT instead transports the selected validation checkpoint A's cut to freshly fitted full-DEV pipeline B, each with its own preprocessor. The supplementary BAF None diagnostic compares A/VAL, A/TEST and B/TEST without selecting a checkpoint or another cut on TEST (Appendix A). B remains the source for all primary results, including transfer and attention; the historical FT network is neither A nor B.

Transfer uses saved Base models, fitted preprocessing and Base DEV-selected thresholds without Variant retraining or recalibration. Base is an in-domain reference, not an upper bound for Variant AP.

Baseline and transfer 95% intervals use 1,000 seed-42 TEST-row bootstrap resamples with fitted models and thresholds fixed; single-class draws are discarded. Primary strategy cells are point estimates. Representation controls use paired resampling with the same rows for both configurations. A separate post-hoc AP comparison uses 1,000 paired seed-42 row resamples for BAF None CatBoost minus LGBM, with aligned frozen scores and percentile bounds. Intervals omit refitting, seed, split and entity-cluster variability. They are not simultaneous guarantees; overlapping marginal intervals or a paired interval containing zero does not establish equivalence. Close transfer values retain four decimals; primary strategy cells retain three. Summaries use complete saved scores and integer confusion counts before display rounding.

3.5. Compatible SHAP and separate attention

BAF SHAP sources are separately pinned to the same frozen LR, LGBM and CatBoost fitted functions used in predictive reporting. LinearExplainer explains LR; TreeExplainer explains the boosted models (Lundberg and Lee, 2017; Lundberg et al., 2020). LR uses an empirical preprocessed Base TEST reference, possibly summarised by its masker; tree explainers use their tree-path-dependent reference. This post-hoc use of unlabelled attributes neither fits the predictive model nor selects its threshold. Different references preclude direct cross-model comparison of absolute attribution magnitudes.

Global original-feature importance uses all 200,000 Base TEST rows. For each observation, contributions are summed with their signs within group G, then the absolute value is averaged: $I_G = \text{mean}_i |\sum_{j \in G} \phi_{ij}|$. This is the

magnitude of the net contribution, not the absolute value of the across-observation mean. The previous encoded-mass summary, $\text{mean}_i \sum_{j \in G} |\phi_{ij}|$, remains a supplementary sensitivity. Signed local grouping preserves additivity. Post-hoc grouping is not a new group-Shapley calculation and does not guarantee encoding invariance. FT-Transformer GradientExplainer is excluded: its wrapper converts floating-point categorical inputs to integer embedding indices, preventing reliable gradient attribution to those inputs.

Variant explanations use the frozen Base-trained LGBM on every retained row. Native tree-path-dependent TreeSHAP operates in raw log-odds and is checked against saved Base TreeExplainer contributions at 128 uniformly spaced TEST positions with absolute tolerance 10^{-10}. Complete importance vectors precede top-10 selection. Across cohorts, $\Delta I_G = I_{G,Variant} - I_{G,Base}$ and $100\Delta I_G / I_{G,Base}$ complement membership with magnitude and rank changes. Relative change is undefined for a zero Base reference. Both policies and all contrasts reuse saved attribution matrices without new SHAP.

Attention uses the baseline BAF FT-Transformer on all 200,000 Base TEST rows. It extracts the final-layer `[CLS]` query row, with PyTorch averaging across heads. Entropy includes all 31 tokens, including `[CLS]`, and is normalised by $\ln 31$. Categorical/numerical attention compares mean weight per token, not total group mass. Diagnostic predictions reproduce saved scores and decisions. Case heatmaps select errors nearest the frozen threshold. Earlier layers and individual heads are outside the aggregate view.

4. Results

4.1. Imbalance sensitivity

Tables 2 and 3 compare FT-Transformer and CatBoost. Population standard deviation across six effective conditions describes intervention spread, not seed variation or significance.

Table 2
FT-Transformer and CatBoost on distinct ULB profiles. Final row: population standard deviation across six effective conditions, calculated before display rounding; † marks SMOTE+Tomek as an observed duplicate of SMOTE, retained in the seven displayed rows.

Strategy	FT AP	FT F_2	CB AP	CB F_2
None	0.801	0.761	0.820	0.775
RUS	0.641	0.600	0.637	0.763
ROS	0.784	0.763	0.811	0.808
SMOTE	0.785	0.692	0.798	0.788
SM+Tomek†	0.785	0.692	0.798	0.788
SMOTEENN	0.797	0.698	0.793	0.779
Weights	0.756	0.774	0.802	0.776
Pop. SD (six)	*0.055*	*0.060*	*0.063*	*0.014*

Table 3
FT-Transformer and CatBoost on BAF Base; same standard-deviation definition as Table 2.

Strategy	FT AP	FT F_2	CB AP	CB F_2
None	0.177	0.304	0.180	0.314
RUS	0.163	0.301	0.178	0.317
ROS	0.166	0.310	0.185	0.316
SMOTE	0.115	0.257	0.154	0.303
SM+Tomek†	0.115	0.257	0.154	0.303
SMOTEENN	0.124	0.269	0.165	0.309
Weights	0.176	0.313	0.184	0.317
Pop. SD (six)	*0.025*	*0.021*	*0.011*	*0.005*

On ULB, RUS produces the largest AP reduction for both models. CatBoost has higher AP except under RUS and SMOTEENN. AP spread is 0.063 for CatBoost versus 0.055 for FT-Transformer, but F_2 spread reverses: 0.014 versus

0.060. Lower dispersion under one metric is not a general robustness advantage. Class Weights reduces transformer AP on ULB while increasing F_2 slightly; the metric and decision rule matter.

BAF baseline AP is 0.1769 for FT-Transformer and 0.1796 for CatBoost, without established equivalence. SMOTE reduces these to 0.1149 and 0.1543: approximately 35% and 14% below baseline. Class Weights is less than 1% below transformer baseline AP and increases F_2 from 0.304 to 0.313, but is not universally safest across datasets. CatBoost ROS reaches 0.185 AP. Figure 1 displays the primary profiles.

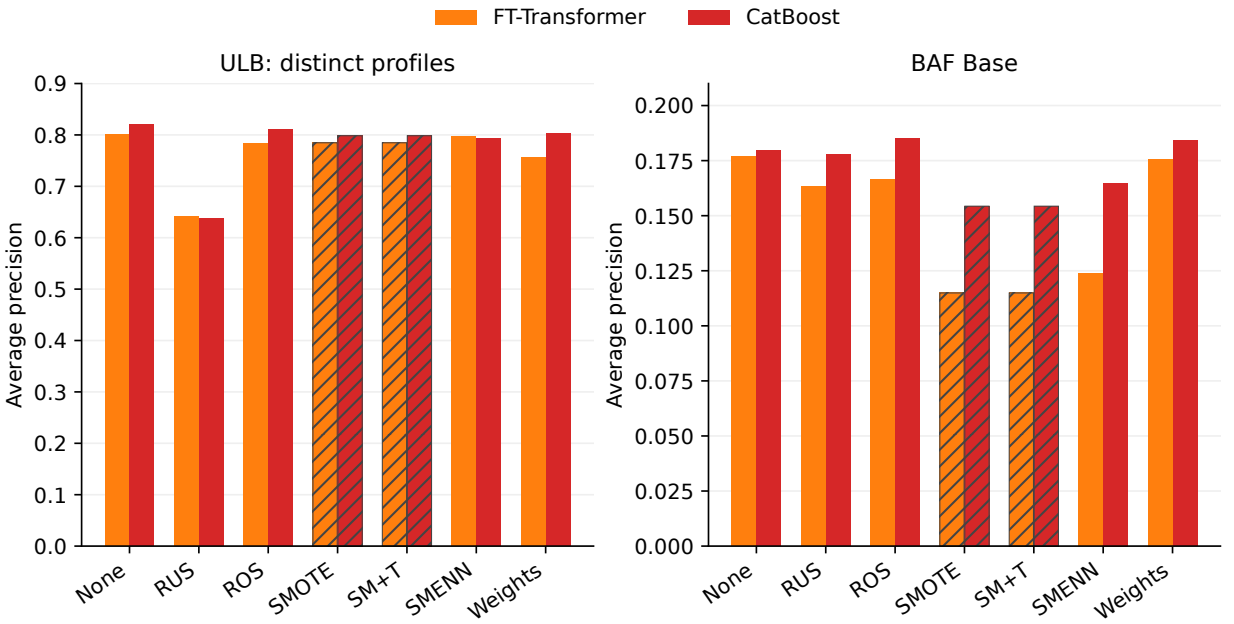


Figure 1: Primary AP point estimates across seven strategies. Hatched bars mark SMOTE and its observed SMOTE+Tomek duplicate. Differences describe implemented sampling representations, not an isolated architectural mechanism.

SMOTE and SMOTE+Tomek are identical effective treatments in logged revised fitting stages: Tomek removes zero rows, with identical inputs and TEST scores. Historical BAF classical scores also coincide up to floating-point noise. A reconstructed full-DEV sampler removes zero rows but does not replace unavailable historical fold logs. This explains equality here, not algorithmic equivalence generally. On BAF, six-condition AP/F_2 spread is 0.025/0.021 for FT-Transformer and 0.011/0.005 for CatBoost. Counting the duplicate once changes these summaries, not any strategy's individual performance.

4.2. Predefined representation controls

Table 4 separates primary SMOTE from the categorical-aware control. Transformer SMOTE-NC reaches AP 0.1157, well below the 0.1769 baseline. The paired change is +0.0008 with interval $[-0.0068, +0.0076]$; it does not establish equivalence to primary SMOTE. Its alert rate changes from 6.11% to 4.22% under separately DEV-selected thresholds, despite similar AP and F_2. CatBoost has lower AP and F_2 in the tested control.

The intervention changes representation and nominal-aware synthesis jointly; it isolates neither architecture nor an optimally tuned alternative. Historical primary CatBoost SMOTE retains its saved six-decimal threshold and reproduced confusion counts. Pairing verifies raw-file identity, seed-42 splitting and identical ordered TEST labels, without assuming an unavailable historical TEST-index artefact.

The absence-code sensitivity has mixed AP directions (Table 5): LR gains 0.0045, while OCSVM loses 0.0028. Remaining AP difference intervals include zero, as do all supervised F_2 difference intervals. These findings neither select an optimal policy nor establish equal policies.

The post-hoc BAF None comparison gives AP CatBoost minus LGBM +0.0028 with paired 95% interval $[-0.0017, +0.0072]$ (Appendix A). The FT SMOTE–None change (−0.0620) and CatBoost SMOTE-NC–SMOTE change (−0.0110) are larger point-estimate differences in these tested pipelines. This is a descriptive contrast of

Table 4
Categorical-aware BAF controls. Differences are SMOTE-NC minus primary SMOTE, with paired 95% row-bootstrap intervals ($B = 1,000$).

Model	Metric	Primary	Control	Δ [paired 95% CI]
CatBoost	AP	0.1543	0.1433	−0.0110 [−0.0147, −0.0074]
	F_2	0.3029	0.2856	−0.0173 [−0.0233, −0.0108]
FT-Transformer	AP	0.1149	0.1157	+0.0008 [−0.0068, +0.0076]
	F_2	0.2567	0.2577	+0.0010 [−0.0088, +0.0111]

Table 5
Absence-code sensitivity. Differences are NaN conversion, mean imputation and indicators minus preserved codes, with paired 95% row-bootstrap intervals ($B = 1,000$).

Model	Metric	Primary	Control	Δ [paired 95% CI]
LR	AP	0.1432	0.1477	+0.0045 [+0.0017, +0.0074]
	F_2	0.2868	0.2910	+0.0042 [−0.0008, +0.0096]
RF	AP	0.1588	0.1596	+0.0008 [−0.0015, +0.0030]
	F_2	0.2941	0.2931	−0.0010 [−0.0061, +0.0044]
LGBM	AP	0.1768	0.1759	−0.0009 [−0.0029, +0.0011]
	F_2	0.3159	0.3173	+0.0015 [−0.0039, +0.0062]
CatBoost	AP	0.1796	0.1790	−0.0005 [−0.0023, +0.0015]
	F_2	0.3143	0.3126	−0.0017 [−0.0063, +0.0031]
FT-Transformer	AP	0.1769	0.1732	−0.0038 [−0.0082, +0.0006]
	F_2	0.3041	0.3026	−0.0016 [−0.0068, +0.0034]
OCSVM	AP	0.0190	0.0162	−0.0028 [−0.0035, −0.0024]
	F_2	0.0805	0.0679	−0.0126 [−0.0166, −0.0087]

magnitudes, not a test of differences between intervention effects or evidence that representation always matters more than model choice.

4.3. Controlled Base-to-Variant transfer

Tables 6–8 report full Base TEST and the same filtered Variants for every model. Bold marks each model's highest cohort value, not the winner within a Variant.

Table 6
Transferred PR-AUC (average precision), without retraining.

Model	Base	Var I	Var II	Var III	Var IV	Var V
LR	0.1432	0.1066	**0.1442**	0.1039	0.1334	0.1005
RF	0.1588	0.1342	**0.1615**	0.1193	0.1477	0.1093
LGBM	0.1768	0.1483	**0.1803**	0.1328	0.1646	0.1185
CatBoost	0.1796	0.1472	**0.1822**	0.1329	0.1668	0.1207
FT-Trans.	**0.1769**	0.1462	0.1751	0.1327	0.1589	0.1168
OCSVM	0.0190	0.0164	**0.0220**	0.0178	0.0205	0.0165

LGBM leads AP on Variant I (0.1483); CatBoost leads Variants II–V (0.1822, 0.1329, 0.1668 and 0.1207). The F_2 leaders differ: CatBoost leads Variant I (0.2701), while LGBM leads II–V (0.3035, 0.2760, 0.3008 and 0.2620). These are point-estimate rankings, not universal or formally established pairwise superiority. RF AP is below Base on four Variants and slightly higher on II (0.1615 versus 0.1588), with no general transfer advantage. All supervised F_2 values fall on every Variant at frozen Base thresholds. The original/filtered audit uses the same fitted function in each

Table 7
Transferred F_2 at frozen Base DEV-selected thresholds.

Model	Base	Var I	Var II	Var III	Var IV	Var V
LR	**0.2868**	0.2391	0.2656	0.2370	0.2680	0.2284
RF	**0.2941**	0.2459	0.2840	0.2522	0.2796	0.2438
LGBM	**0.3159**	0.2693	0.3035	0.2760	0.3008	0.2620
CatBoost	**0.3143**	0.2701	0.2994	0.2697	0.2991	0.2588
FT-Trans.	**0.3041**	0.2661	0.2655	0.2489	0.2722	0.2391
OCSVM	0.0805	0.0662	**0.0925**	0.0749	0.0870	0.0690

Table 8
Transferred ROC-AUC under the common 30-feature projection.

Model	Base	Var I	Var II	Var III	Var IV	Var V
LR	0.8745	0.8488	0.8786	0.8517	**0.8819**	0.8450
RF	0.8778	0.8533	0.8851	0.8586	**0.8880**	0.8461
LGBM	0.8985	0.8820	0.9004	0.8809	**0.9062**	0.8710
CatBoost	0.8980	0.8812	0.8994	0.8797	**0.9042**	0.8712
FT-Trans.	0.8965	0.8785	0.8964	0.8760	**0.9009**	0.8681
OCSVM	0.6279	0.5959	**0.6616**	0.6144	0.6456	0.5907

pair, including B for FT. With repeated Base DEV profiles included, RF AP is 39–53% above its filtered value and ranks first rather than fourth. RF detects all excluded frauds at its frozen cut, consistent with benefit from previously seen profiles; changing composition and prevalence prevents a unique causal memorisation claim (Appendix C). Under Base-to-filtered-Variant transfer, supervised alert rates fall by 12–16% on I and rise by 50–64% on II–V. Frozen cuts therefore do not preserve review rates or budgets; these are rate changes, not equal count changes on the smaller Variant cohorts.

Variant II AP is slightly above Base for the four classical supervised models and slightly below for FT-Transformer. An increase is compatible with verified profile separation: ranking and population can change in either direction. Variants III/V produce approximately 25–34% reductions for the leading Base models, but omitted separability variables, pooled months and filtering prevent an isolated causal explanation. Supervised Variant ROC-AUC spans approximately 0.845–0.906. ROC-AUC is insensitive to pure class-prior changes with fixed class-conditional score distributions, but those distributions may change here. OCSVM stays below 0.023 AP on each Variant; this bounds its performance here, not anomaly detection generally.

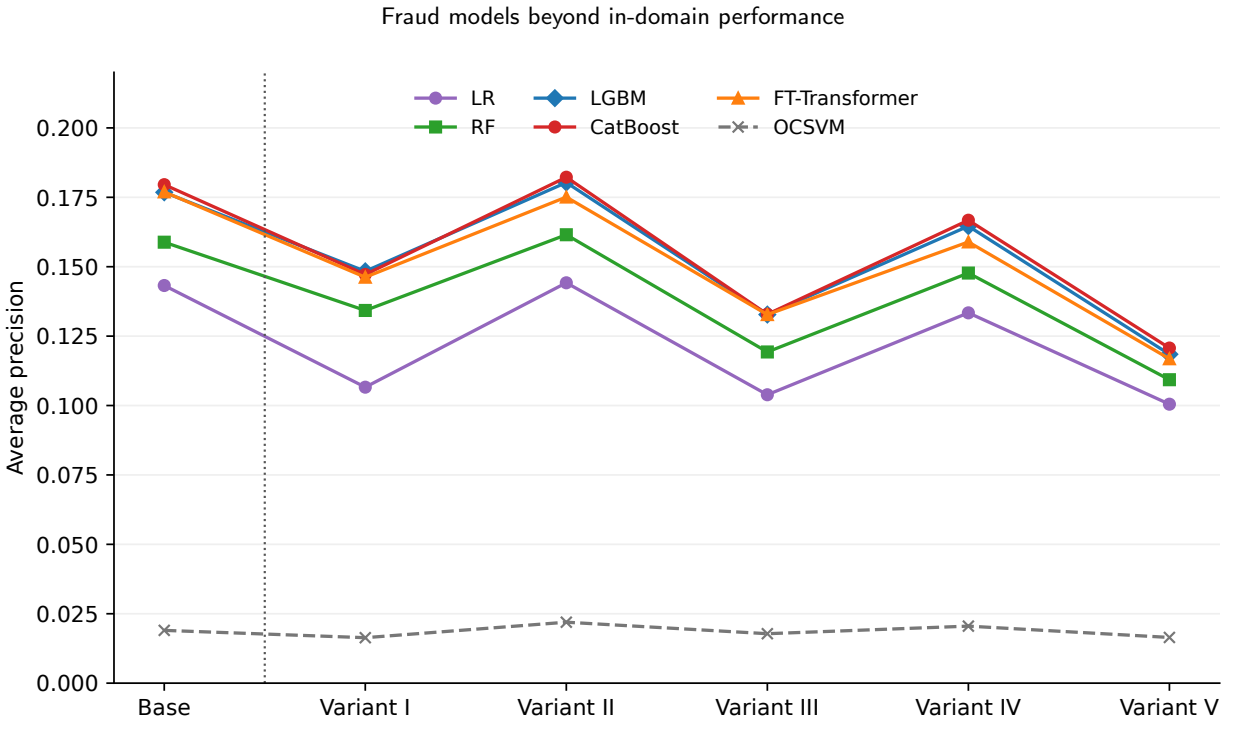


Figure 2: Base and retained-Variant AP. The dashed separator distinguishes in-domain evaluation from controlled transfer. LGBM leads Variant I and CatBoost leads II–V by point estimate.

4.4. Compatible SHAP feature sets

Figure 3 shows each compatible model's own top 15 original features. `device_os` and `phone_home_valid` are among each model's leading three, but the trios differ: LR and CatBoost also include `has_other_cards`; LGBM includes `housing_status`. Absolute magnitudes are not directly comparable across explainer references.

LGBM and CatBoost share all top-ten members (Jaccard 1.00). Each shares eight with LR, giving $8/(10+10-8) = 0.67$ (Table 9).

Table 9
Cross-model Jaccard agreement of top-10 original-feature SHAP sets.

Model	LR	LGBM	CatBoost
LR	1.00	0.67	0.67
LGBM	0.67	1.00	1.00
CatBoost	0.67	1.00	1.00

The fixed Base-trained LGBM retains the same top-10 set across all cohorts (Table 10). Magnitudes nevertheless change beyond that set: `customer_age` increases from 0.1081 on Base to 0.1851–0.1856 log-odds on II–V (71–72%, rank 17 to 11), but decreases 15.9% on I (rank 18). The DOB distinct-email feature rises by 33.5–34.6% on II–V and falls by 7.2% on I. Both are single-column numerical variables, so their magnitude changes do not arise from category regrouping. These are population contrasts for one fixed model, not new learning or equivalent percentage changes in fraud probability.

The complete signed decomposition preserves additivity; the residual contains all omitted contributions. These deliberately selected extremes illustrate model behaviour, not representative error-class estimates. A false-positive explanation describes a high score without validating it. The lowest-score FN is not borderline, and these examples do not quantify the workload needed to recover all missed frauds.

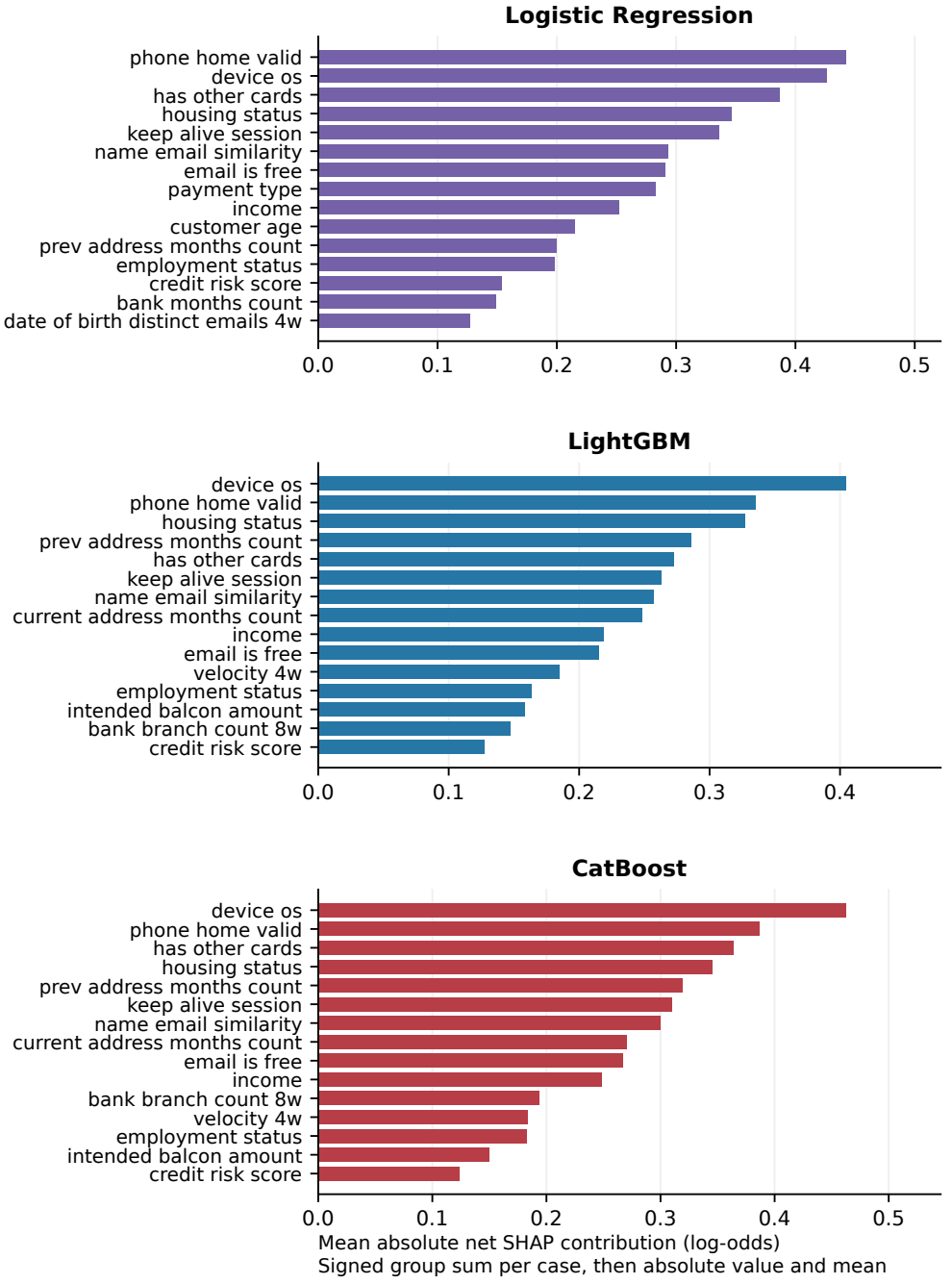


Figure 3: Each compatible model's own top-15 original-feature importance, in log-odds. Signed one-hot contributions are summed per observation before absolute value and averaging. FT-Transformer gradient SHAP is excluded.

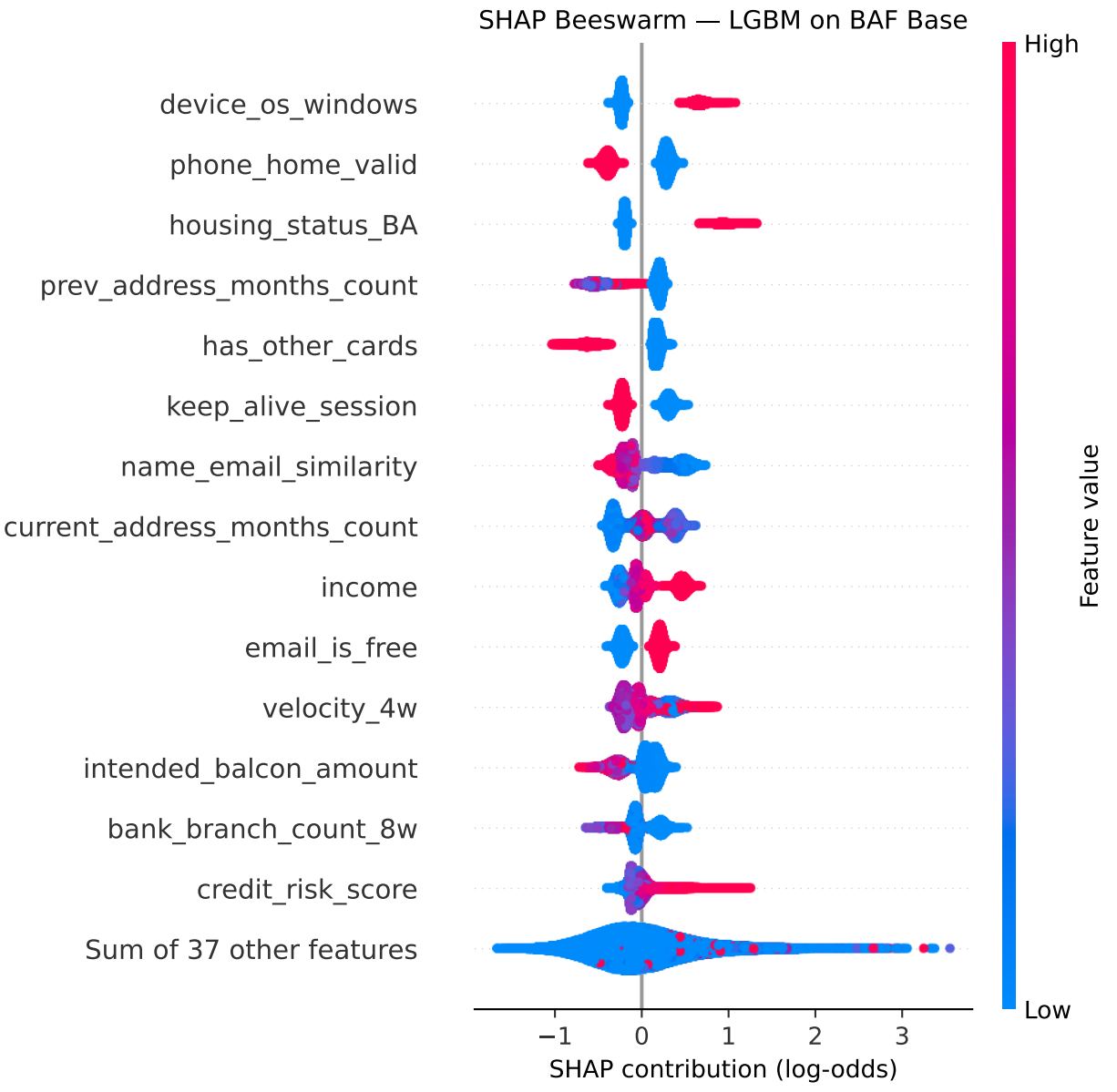


Figure 4: LGBM SHAP beeswarm in log-odds: 14 leading encoded columns and the signed sum of the remaining 37. Colour denotes transformed feature values, not causal fraud effects.

Table 10
Jaccard agreement for frozen LGBM top-10 original-feature sets on Base TEST and filtered Variant cohorts.

Cohort	Base	I	II	III	IV	V
Base	1.00	1.00	1.00	1.00	1.00	1.00
I	1.00	1.00	1.00	1.00	1.00	1.00
II	1.00	1.00	1.00	1.00	1.00	1.00
III	1.00	1.00	1.00	1.00	1.00	1.00
IV	1.00	1.00	1.00	1.00	1.00	1.00
V	1.00	1.00	1.00	1.00	1.00	1.00

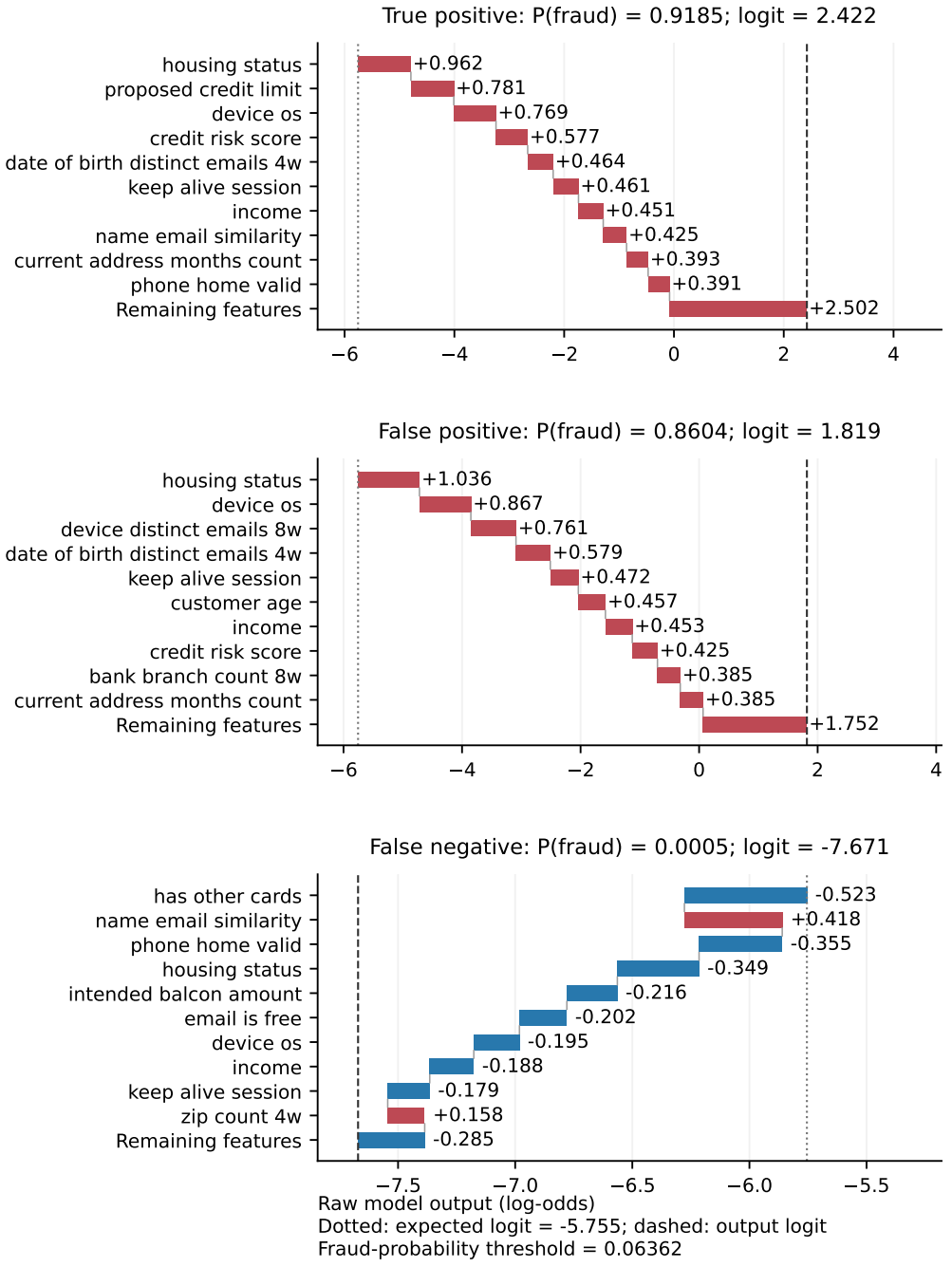


Figure 5: Additive LGBM log-odds waterfalls for highest-score TP and FP and lowest-score FN. Ten signed original-feature contributions plus the residual connect expected and case outputs; red increases the output towards fraud.

4.5. Separate FT-Transformer attention diagnostics

Mean normalised entropy is 0.985 (mean entropy 3.381 nats; $\ln 31 = 3.434$). Mean [CLS] self-attention is 3.6%, compared with a uniform reference of about 3.2%. Mean weight per categorical/numerical token is 4.0%/3.1%; the unrounded ratio is 1.311. These descriptive statistics do not define formal pathology thresholds.

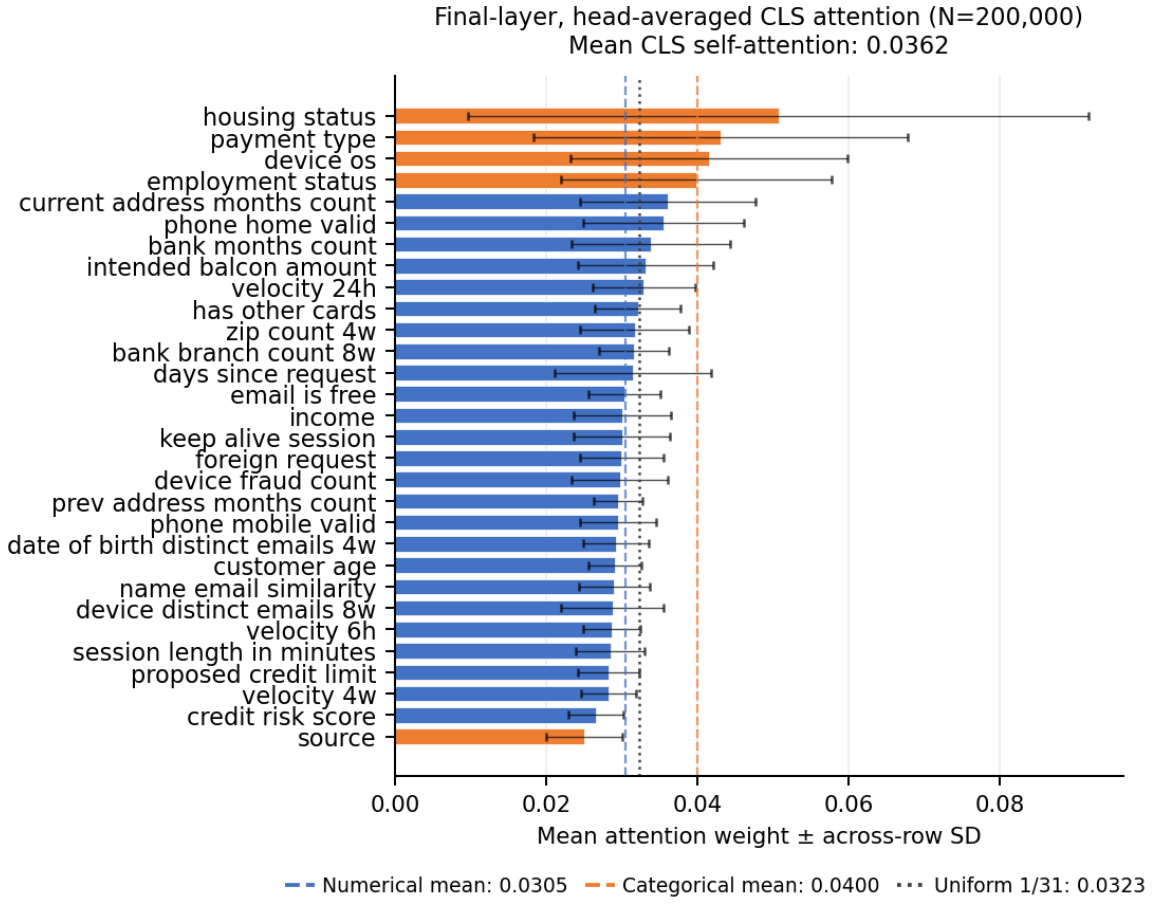


Figure 6: Final-layer, head-averaged [CLS] attention on 200,000 BAF Base TEST rows. Error bars are standard deviations across samples; uniform reference: 1/31. Group means are not total group mass.

Largest mean feature-token weights are `housing_status` (5.1%), `payment_type` (4.3%) and `device_os` (4.2%). Aggregate [CLS] attention is diffuse; individual queries, heads and earlier layers need not be. Housing status and device operating system also appear among leading SHAP features, but overlap establishes neither faithful attention attribution nor cross-method causal validity.

The matrix has an elevated housing-status key column for several queries. This selected error does not show uniform diffusion for every query or error. Diagnostic scores have zero threshold disagreements over 200,000 TEST rows, verifying extraction fidelity, not explanatory faithfulness. The weights neither explain close baseline AP values nor bound possible improvements from another model, representation or feature set.

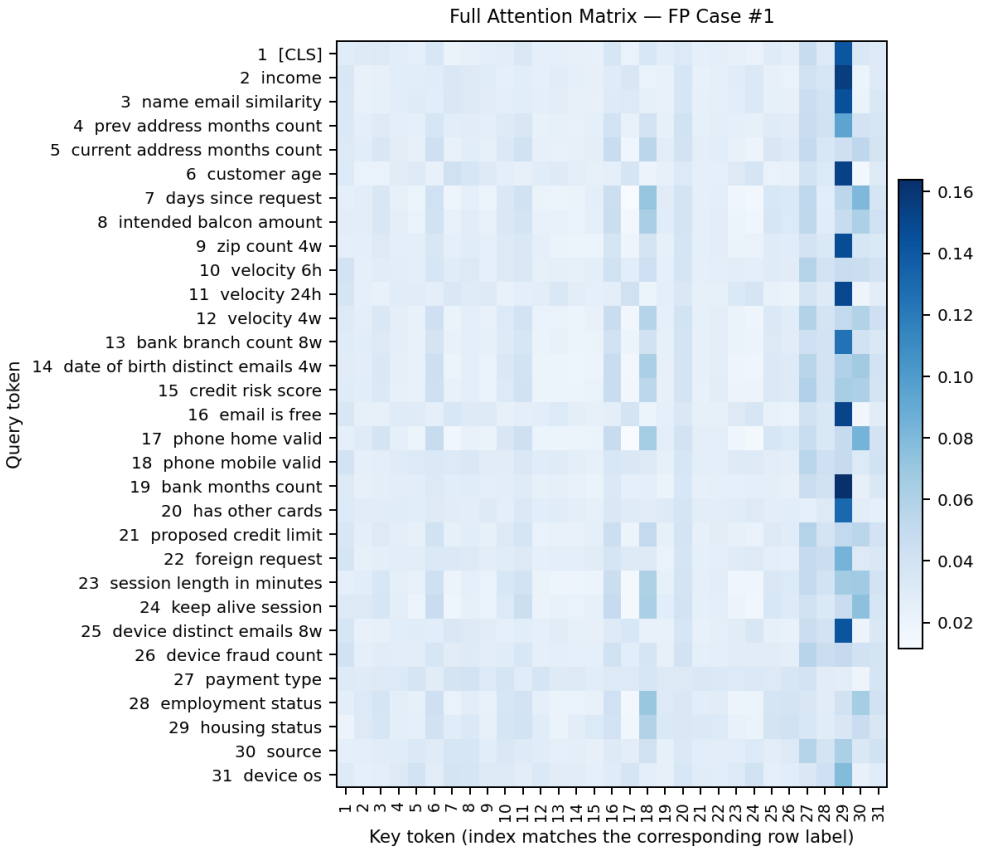


Figure 7: Final-layer, head-averaged 31×31 matrix for the FP nearest the frozen threshold ($p \approx \tau = 0.0584$). Cells are query-to-key weights; case-specific elevations need not resemble aggregate `[CLS]` attention.

4.6. Recorded computational burden

Table 11
Recorded BAF baseline costs (seconds). †: historical measurements retained with frozen classical models. Revised FT-Transformer has no new search; validation recovery is separate.

Model	Tuning (s)	Train (s)	Inference (s)
LR	55,070†	83.9†	0.209
RF	105,313†	1494.0†	10.642
LGBM	1,243†	16.1†	1.298
CatBoost	7,010†	36.7†	0.213
FT-Trans.	—	354.4	4.262
OCSVM	—	919.6†	71.579†

LGBM has lower recorded final-fit/inference times than FT-Transformer: 16.1/1.298 versus 354.4/4.262 seconds. Classical models execute on CPU and the transformer on GPU on an i5-13600KF/16 GB/RTX 3070 machine. Historical/revised executions span Python 3.12.7/3.12.14 and model-specific validation work; concurrent workloads and realised fit lengths also matter. There is no new BAF transformer search time comparable with retained classical tuning times. Records describe resource burden, not an intrinsic speed ratio or matched-hardware benchmark.

5. Discussion

5.1. Model and representation are jointly evaluated

The larger transformer SMOTE reduction on BAF is a pipeline difference. Both primary paths alter encoded features using different category representations. Integer conversion of interpolated ordinal values is not interpolation of learned embeddings; trees also respond to altered training representations. The categorical-aware control does not recover baseline transformer AP, but changes representation and synthesis jointly, retains fixed hyperparameters and leaves numerical binary constraints unenforced. It cannot identify a unique architectural failure.

Absence-code sensitivity jointly changes conversion, imputation and indicators. Its mixed effects justify explicit representation reporting, not a universal policy. The primary policy stays predefined. Weights preserve training rows, not necessarily the original-prevalence score scale. A supplementary analytic prior/weight transformation reduces Brier in all 15 BAF RUS/ROS/Weights cases without replacing primary scores or cuts (Appendix B). Improved probabilistic error is not proof of calibration; the transformation also does not resolve transport between validation and final models. Class Weights results likewise do not establish an untested objective, such as focal loss, as an improvement. New objectives and independently tuned categorical-aware pipelines remain future-work candidates.

5.2. Transfer depends on metric and population

LGBM and CatBoost lead transferred point estimates, with the leader depending on metric and Variant. That is not a generic advantage under every shift. The synthetic populations, pooled months, omitted separability features and profile filtering define the scope.

Every supervised F_2 falls at frozen thresholds even where AP rises. Preserved ranking quality therefore does not guarantee preserved operational performance. Freezing Base thresholds is the experiment, not proof that they are optimal or stable on every Variant. Prospective use would require appropriate temporal or external evaluation and a separately validated decision rule.

5.3. Interpretation agreement is bounded

Compatible global SHAP sets provide common candidate features for inspection. Stable top-10 membership for fixed LGBM is useful descriptive evidence, not a waiver of monitoring, local explanation review or operational revalidation. Neither common membership nor additivity establishes causality.

The attention diagnostic asks how concentrated one final-layer head-averaged allocation is. Diffusion is not evidence that the inputs contain no additional signal. It cannot identify an information-theoretic AP ceiling, diagnose all pathological behaviour or rule out improvements from other architectures. Comparing model reliance would require compatible output-sensitive methods and validation of their relationship to predictions.

5.4. Limitations and reproducibility

The study uses one outer split and saved fitted models, not repeated seeds. Classical five-fold validation and transformer holdout estimate different selection variability. Equal attempted trial budgets do not imply equal compute or search coverage; strategies are not independently retuned.

ULB distinct-profile estimates do not concern its original transaction-frequency population; PCA features limit domain interpretation. BAF's shared synthetic process and possible TEST overlap limit independent deployment inference. Exact-profile safeguards do not prove entity independence. Row-bootstrap intervals omit training and cluster variability; multiple comparisons are not adjusted as simultaneous guarantees.

Interpretation depends on models, explainer references, grouping, top-k choice and attention aggregation. The custom transformer is the sole deep-learning representative. Mixed execution provenance prevents intrinsic cost comparison.

A dated manifest pins configurations, scores, indices, thresholds, transfer cohorts, compatible explanations and reporting assets. Article tables are extracted from verified outputs; plots read saved evidence without fitting. Traceability complements, rather than substitutes for, independent empirical replication.

6. Conclusion

Tree ensembles remain strong candidates in this setting, but the evidence supports conditional comparison rather than universal sufficiency. Primary SMOTE has a larger BAF AP reduction for FT-Transformer than CatBoost; a fixed categorical-aware control does not restore its baseline. Representation remains a confound.

After exact-profile filtering, LGBM leads AP on Variant I and CatBoost on II–V. Their F_2 leadership differs by Variant, and all supervised models lose F_2 at frozen thresholds. Compatible SHAP comparisons establish retained global membership alongside changing contribution magnitudes, not identical rankings or deployment-ready explanations. For fixed LGBM, mean age-contribution magnitude increases 71–72% on II–V relative to Base. Diffuse final-layer attention is a separate diagnostic, not exhausted signal.

The practical contribution is a traceable model–representation–threshold comparison on controlled populations. Future work should examine repeated seeds, temporal/external validation, independently tuned categorical-aware pipelines, earlier layers and individual heads, and compatible output-sensitive explanations. These extensions test unresolved questions without presuming an architectural mechanism or data-imposed ceiling.

A. Supplementary operating-point diagnostics

All diagnostics retain the primary fitted models and cuts. In the BAF None FT sensitivity, A denotes the selected validation checkpoint and its own preprocessor; B denotes the full-DEV final pipeline. A reproduces archived VAL scores bit for bit before TEST evaluation. A/VAL contains 160,000 cases; A/TEST and B/TEST use the same 200,000. B/VAL is not independent validation. No checkpoint is selected by these TEST comparisons.

Table 12
BAF None FT transport at the same original max-F_2 cut, 0.05843411013484001. P and R denote precision and recall; B/TEST remains primary.

Pipeline/cohort	AP	P	R	F_2	Alerts	Rate (%)
A/VAL	0.1707	0.132	0.468	0.310	6245	3.903%
A/TEST	0.1767	0.136	0.475	0.317	7710	3.855%
B/TEST	0.1769	0.108	0.558	0.304	11414	5.707%

The same-TEST contrast changes fitted weights and preprocessing jointly. Close AP accompanies different recall and workload; it does not isolate a cause or justify replacing B by A. Classical fold cuts are summarised below: the median limits extreme-cut influence but need not optimise the final fit. FT uses a single holdout, so fold dispersion is not applicable.

Table 13
Baseline DEV-selected max-F_2 cuts and fraud counts per validation partition. N/A denotes non-applicable between-fold dispersion; OCSVM uses an anomaly-score scale. Full cuts are retained in the supplementary evidence.

Dataset	Model	Frauds/ fold	Min.	Median/ holdout	Max.
ULB	LR	75–76	0.011198	0.055033	0.094248
ULB	RF	75–76	0.130000	0.260000	0.410444
ULB	LGBM	75–76	0.068431	0.158063	0.424815
ULB	CatBoost	75–76	0.081978	0.127928	0.248675
ULB	FT-Trans.	76 (holdout)	N/A	0.273683	N/A
ULB	OCSVM	75–76	1.412886	2.239141	2.399230
BAF	LR	1764–1765	0.045109	0.052715	0.059991
BAF	RF	1764–1765	0.060521	0.063114	0.068504
BAF	LGBM	1764–1765	0.053179	0.063621	0.071221
BAF	CatBoost	1764–1765	0.055428	0.057330	0.072526
BAF	FT-Trans.	1765 (holdout)	N/A	0.058434	N/A
BAF	OCSVM	1764–1765	-43.165660	-40.259704	-35.650378

For the CatBoost minus LGBM AP contrast in Table 14, the same row indices are resampled for the two frozen score arrays after checking labels and ordered row identities. Single-class draws are excluded. This interval omits refitting and entity-cluster uncertainty and does not establish equivalence.

Table 14
Post-hoc BAF None AP contrast. Paired TEST-row bootstrap: 1,000 valid draws of 1,000 requested, NumPy PCG64 seed 42, percentile interval.

Contrast (BAF None)	Δ AP	Paired 95% interval
CatBoost − LGBM	+0.0028	[-0.0017, +0.0072]

B. Supplementary prior/weight score-scale sensitivity

For the five BAF supervised models, None is the original reference and RUS/ROS/Weights supply 15 transformations. Frozen scores are mapped by $g(s) = cs/[cs+(1-s)]$. RUS/ROS use $c = [\pi/(1-\pi)](1-q)/q$, with original DEV prevalence $\pi = 8,823/800,000$ and configured balanced final-training proportion $q = 0.5$. The prior interpretation assumes suitable posterior scores and unchanged within-class distributions (Saerens, Latinne and Decaestecker, 2002). Weights use $c = w_0/w_1$ as a weighted-posterior sensitivity. No TEST-derived prior or iterative calibration procedure is estimated. FT final-fit logs confirm the sampled counts; historical classical full-DEV reconstruction does not replace unavailable original fold sampler logs.

Table 15
BAF frozen-score sensitivity. None is untransformed. Cuts are available source cuts displayed to six decimals; count replay checks archived confusion counts without TEST threshold search.

Model	Strategy	Raw Brier	Adjusted Brier	τ raw	$g(\tau)$	CM replay
LR	None	0.010091	0.010091	0.052715	0.052715	Yes
LR	Rus	0.141427	0.010128	0.835949	0.053770	Yes
LR	Ros	0.140332	0.010130	0.847697	0.058442	Yes
LR	Weights	0.112924	0.010331	0.862054	0.065149	Yes
RF	None	0.010067	0.010067	0.063114	0.063114	Yes
RF	Rus	0.142963	0.010297	0.753884	0.033031	Yes
RF	Ros	0.013206	0.010921	0.206966	0.002902	Yes
RF	Weights	0.013559	0.010921	0.210354	0.002962	Yes
LGBM	None	0.009864	0.009864	0.063621	0.063621	Yes
LGBM	Rus	0.129658	0.009905	0.851913	0.060286	Yes
LGBM	Ros	0.115265	0.009900	0.850052	0.059460	Yes
LGBM	Weights	0.115342	0.009901	0.839609	0.055157	Yes
CatBoost	None	0.009843	0.009843	0.057330	0.057330	Yes
CatBoost	Rus	0.128474	0.009854	0.849720	0.059315	Yes
CatBoost	Ros	0.117131	0.009820	0.836854	0.054108	Yes
CatBoost	Weights	0.118106	0.009821	0.826635	0.050489	Yes
FT-Trans.	None	0.009959	0.009959	0.058434	0.058434	Yes
FT-Trans.	Rus	0.146374	0.010211	0.921788	0.116164	Yes
FT-Trans.	Ros	0.119837	0.009938	0.854138	0.061299	Yes
FT-Trans.	Weights	0.103967	0.009898	0.841275	0.055808	Yes

All 15 cases reduce Brier. Transforming scores and available cuts preserves decisions, and all 12 historical classical six-decimal cuts reproduce their archived counts. Exact arithmetic preserves ordering; float64 coalesces 16 RF–ROS and 536 RF–Weights adjacent distinct-score pairs, without reversals or decision changes. RF–Weights AP changes by -1.22×10^{-9} and ROC-AUC by $+4.93 \times 10^{-7}$. Complete checks and metrics accompany the diagnostic. This is not guaranteed calibration or proof that rebalance effects are solely rescaling. Synthetic samplers and OCSVM are excluded, and primary scores, transfer, SHAP and attention remain untransformed.

C. Supplementary transfer and attribution evidence

Original/filtered comparisons hold each fitted function fixed, including FT B. They quantify overlap-associated differences rather than an independent transfer result or a comparison with the historical FT network.

Table 16
RF overlap diagnostic. Relative AP is $100(AP_{original}/AP_{filtered} - 1)$; ranks compare all six models.

Variant	Excluded	AP original	AP filtered	Relative (%)	Rank
I	22,474	0.2058	0.1342	+53.3	1 to 4
II	22,469	0.2245	0.1615	+39.0	1 to 4
III	21,282	0.1699	0.1193	+42.4	1 to 4
IV	21,350	0.2108	0.1477	+42.7	1 to 4
V	21,172	0.1593	0.1093	+45.8	1 to 4

Table 17
Base and filtered-Variant alert rates (%) at fixed Base cuts. This comparison differs from original/filtered overlap contrasts.

Model	Base	I	II	III	IV	V
LR	4.14	3.48	6.74	6.76	6.75	6.79
RF	3.36	2.88	5.40	5.46	5.47	5.46
LGBM	3.25	2.80	5.09	5.12	5.08	5.08
CatBoost	3.80	3.23	5.95	5.96	5.95	5.99
FT-Trans.	5.71	5.01	8.59	8.56	8.56	8.58
OCSVM	7.32	6.85	9.33	8.92	8.90	8.98

Table 18
Mean net-contribution magnitude in log-odds, differences from Base and ranks among all 30 LGBM variables. DOB-email denotes `date_of_birth_distinct_emails_4w`.

Cohort	Mean magnitude	Δ magnitude	Relative (%)	Rank
Customer age				
Base	0.1081	+0.0000	+0.00	17
I	0.0909	-0.0172	-15.87	18
II	0.1856	+0.0775	+71.70	11
III	0.1855	+0.0775	+71.66	11
IV	0.1851	+0.0770	+71.27	11
V	0.1851	+0.0770	+71.25	11
DOB distinct emails (4 weeks)				
Base	0.0814	+0.0000	+0.00	20
I	0.0755	-0.0059	-7.23	20
II	0.1096	+0.0282	+34.62	18
III	0.1090	+0.0275	+33.81	18
IV	0.1090	+0.0275	+33.82	18
V	0.1087	+0.0273	+33.50	18

The accompanying machine-readable supplement supplies all 30 same-model overlap pairs in `transfer_overlap_pairs.csv`, cohort sizes and alert counts in `transfer_alert_workload.csv`, and all 30-feature LGBM magnitude/rank contrasts in `shap_magnitude_contrasts.csv`. The three `shap_aggregation_logreg.csv`, `shap_aggregation_lgbm.csv`, and `shap_aggregation_catboost.csv` vectors retain both the main net magnitude and encoded-mass sensitivity. Provenance identifies the saved attribution inputs, common cohorts, model identities and aggregation policy; no new SHAP or model fitting is involved.

CRediT authorship contribution statement

Olavo Caixeiro: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft.
Maryam Abbasi: Conceptualization, Supervision, Writing – review & editing.
Pedro Miguel de Oliveira Martins: Supervision, Writing – review & editing.

Data Availability

ULB is available from Machine Learning Group, Université Libre de Bruxelles (2016) and BAF from Feedzai Research (2022), subject to their respective terms. The research code, article-specific aggregate results and dataset access instructions are available in the dedicated repository at https://github.com/Applied-Intelligence-Hub/transformers-vs-ml-fraud-detection. The wider study and the dissertation manuscript are available at https://github.com/Olavo200100274/fraud-imbalance-classical-vs-transformers. Corrected evidence is identified by the dated 20261005 manifest. The subsequent supervisor-revision derivatives separately identify the expected top-k policy, six-condition summaries, net SHAP aggregation and post-hoc diagnostics, without overwriting the original experimental archive. The dedicated release manifest specifies the distributed frozen predictions, aggregate results and plot-ready explanation inputs. These support numerical and graphical regeneration without refitting. Trained model weights and the complete attribution and training archives are not distributed; the package does not claim exact replay of every historical fitted function. The article PDF remains an unpublished manuscript. No publication or acceptance status is asserted for the companion manuscript.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

OpenAI Codex was used to assist with manuscript revision, checks against the saved experimental evidence, and reproducible plotting and layout code for the figures. Experimental values were read from recorded evaluations, not invented or inferred by a generative model. The authors remain responsible for the accuracy, interpretation and final approval of the manuscript.

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